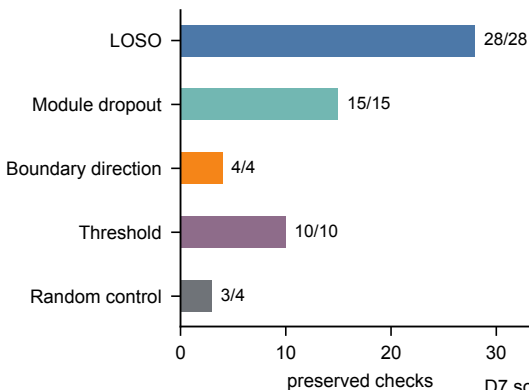
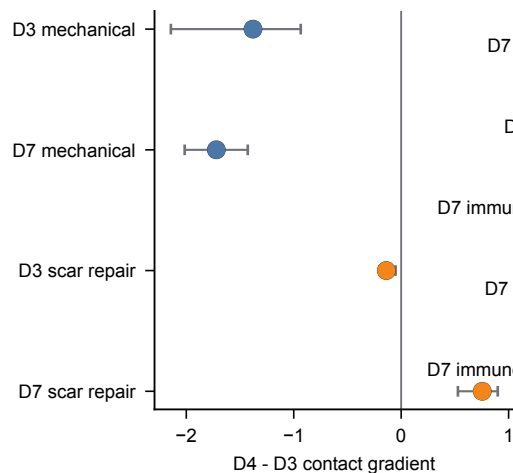


b**a**

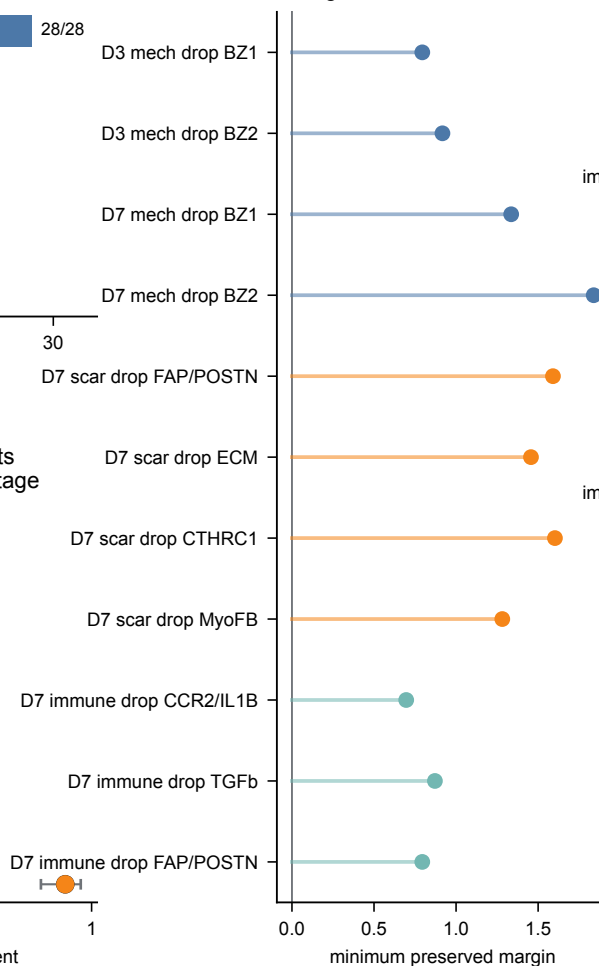
Public-data robustness

**c**

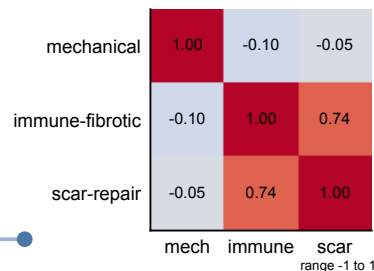
Boundary gradients change with repair stage



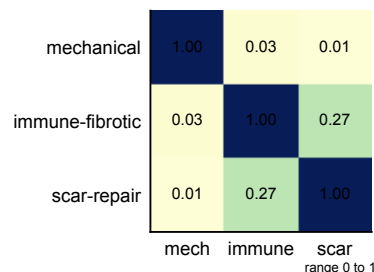
No single module drives the axis call

**d**

Spearman r



Top-decile Jaccard



Human STEMI transfer summary

Mechanical-border and fibroblast-scar axes exceeded random controls. The immune-fibrotic axis was directionally positive but treated as a contextual coupling signal. Human STEMI maps showed distinct mechanical hotspots and partial immune-scar overlap, supporting three outputs.